

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)
DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL3–MCQTest

CourseCode:	CSA0305	CourseTitle:	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	Assessment:	AssessmentTool3–MCQTest
Weightage:	30%ofCIA	TotalMarks:	100
CO3Statement:	SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	H SARAVANA	Reg. No.:	192511397
Section/Batch:	CSE	Date:	12/08/2026

CodeofConduct

I H SARAVANA (192511397)(Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature ofthe Student: H saravana

Instructions

- Thistestcontains25MCQTestquestions(4markseach),Total:100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements:
Identifythemostappropriateanswer,conceptualunderstanding,problem-solvingability,applicationof knowledge.
- Nonenegativemarking.Unansweredquestionscarryzeromarks.

Section/Topic	Focus	QNos	Marks
Preliminaries–TreeTerminology	Root,height,level,siblingsandsubtree	1–2	8
BinaryTree, BinarySearchTree	Properties,binarytreeandbinarysearchtree,in sersion and deletion	3–9	28
Tree Traversals	Traversaltypes,Inorder,PostorderandPreorder	10–14	20
AVL Tree	Balancingfactor,Rotation,Typesofrotation,examples	15–16	8
ApplicationsofSearchTrees	Usageofbinarytree	17	4
BTree -2-3tree,2-3-4tree	Properties,ExampleandOperations	18–19	8
TRIE	Properties,structuresandOperations	20	4
RedBlackTree	Properties,BalancingandOperations	21–22	8
SplayTree	Splaying,Typesandexamplesforinsertionanddeletion	23–25	12
GRANDTOTAL:			100Marks

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrate thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Show good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem-based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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QUESTIONS

Q1

Tree Traversal

4 marks

**Q2**

Binary Search Tree

4 marks

**Q3**

Red-Black Tree

4 marks

**Q4**

Red-Black Tree

4 marks

**Q5**

B-Tree

4 marks

**Q6**

Tree Traversal

**Q1 Tree Traversal**

4 marks

You analyze the post-order traversal of a tree representing project tasks: [D, E, B, F, G, C, A]. You want to know

A. 3**B.** 4**C.** 5**D.** 2 Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q2 Binary Search Tree

4 marks

In a Binary Search Tree, the left child contains:

A. Greater values

B. Smaller values

C. Equal values

D. Random values

 Save Answer

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QUESTIONS

Q1Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6****Q3 Red-Black Tree**

4 marks

During insertion, when a red node has a red parent and the uncle is also red, what action is taken:

A. Single rotation**B.** Double rotation**C.** Recolor parent and uncle black, grandparent red**D.** No action needed Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q4 Red-Black Tree

4 marks

Which property ensures that no two red nodes are adjacent in a Red-Black Tree?

A. Root is black

B. Red nodes have black children

C. Every path from root to leaf has the same number of black nodes

D. Leaves are red

Save Answer

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QUESTIONS

Q1Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6****Q5 B-Tree**

4 marks

A 2-3-4 tree contains a 4-node with keys [20, 40, 60]. If a new key 50 is to be inserted, what is the result of the split?

A. New root becomes [40]; children: [20], [50], [60]

B. Root stays [20, 40, 50, 60]

C. Keys rearranged into a single 4-node

D. Insertion is not possible

 Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6



Q6 Tree Terminology

4 marks

A subtree of a tree is:

A. A separate graph

B. A part of the tree that is itself a tree

C. Only leaf nodes

D. Only internal nodes



Save Answer

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QUESTIONS

Q1

Tree Traversal

4 marks



Q2

Binary Search Tree

4 marks



Q3

Red-Black Tree

4 marks



Q4

Red-Black Tree

4 marks



Q5

B-Tree

4 marks



Q6



Q7 Tree Terminology

4 marks

The children of the same parent are called:

A. Ancestors

B. Descendants

C. Siblings

D. Leaves



Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6



Q8 Binary Search Tree

4 marks

How many distinct binary search trees can be created out of 4 distinct keys?

A. 4

B. 42

C. 24

D. 14

 Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q9 Binary Tree

4 marks

The left and right subtrees of a binary tree are themselves:

A. Trees

B. Graphs

C. Arrays

D. Linked Lists

 Save Answer

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QUESTIONS

Q1Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q10 Binary Tree**

4 marks

A complete binary tree is one in which:

- A.** All levels are fully filled
- B.** Every node has two children
- C.** All levels except possibly the last are completely filled
- D.** Only root is filled

 Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6



Q11 Binary Tree

4 marks

The maximum number of nodes at level 'i' in a binary tree is:

A. 2^i

B. i

C. i^2

D. $2i$

Save Answer

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QUESTIONS

Q1Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6**

Tree Terminology

**Q12 Binary Search Tree**

4 marks

In a BST managing product SKUs, values 20, 15, 25, 10, 18, 30 are inserted. To optimize retrieval, you check the number of edges between node 10 and the root. How many edges are in the path from node 10 to the root?

A. 1**B.** 3**C.** 2**D.** 5 Save Answer

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CO3_AT3_MCQ[← Back](#)**QUESTIONS****Q1**Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6**

Tree Terminology

**Q13 Tree Traversals**

4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: In-order and post-order. Given: In-order: D B E A F C Post-order: D E B F C A What is the root

A. A**B.** D**C.** B**D.** C **Save Answer****Next →**



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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6

Tree Terminology
4 marks



Q7

Tree Terminology



Q14 Tree Traversals

4 marks

Which traversal follows Left → Root → Right?

- a)
- b)
- c)
- d)

A. Level order

B. Preorder

C. Postorder

D. Inorder



Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6

Tree Terminology
4 marks



Q15 Tree Traversals

4 marks

Preorder traversal of an expression tree gives

A. Prefix expression

B. Infix expression

C. Postfix expression

D. Sorted expression

 Save Answer

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QUESTIONS

Q1

Tree Traversal

4 marks



Q2

Binary Search Tree

4 marks



Q3

Red-Black Tree

4 marks



Q4

Red-Black Tree

4 marks



Q5

B-Tree

4 marks



Q6

Tree Terminology

4 marks



Q7

Tree Terminology

4 marks



Q16 Tree Traversals

4 marks

Consider the binary tree:

10

/ \

20 30

/ \ / \

40 50 60 70

What is the Preorder traversal?

A. 40 20 50 10 60 30 70

B. 40 50 20 60 70 30 10

C. 10 20 40 50 30 60 70

D. 10 30 60 70 20 40 50

 Save Answer

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QUESTIONS

Q1Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6**Tree Terminology
4 marks**Q7****Q17 AVL Tree**

4 marks

The balance factor of an AVL tree node can be:

A. -1, 0, +1**B.** -2, -1, 0**C.** 0, 1, 2**D.** Any value Save Answer

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QUESTIONS

Q1

Tree Traversal

4 marks

**Q2**

Binary Search Tree

4 marks

**Q3**

Red-Black Tree

4 marks

**Q4**

Red-Black Tree

4 marks

**Q5**

B-Tree

4 marks

**Q6**

Tree Terminology

4 marks

**Q7**

Tree Terminology

**Q18 AVL Tree**

4 marks

Consider the AVL tree: What is the balance factor of node 30?

30

/\

20 40

/\

10 25

B. +1**C.** -1**D.** +2 Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6

Tree Terminology
4 marks



Q19 Search Tree

4 marks

Which of the following is a common application of Binary Search Trees (BST)?

A. Database indexing

B. Video editing

C. Audio compression

D. Network wiring

 Save Answer

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QUESTIONS

Q1

Tree Traversal

4 marks



Q2

Binary Search Tree

4 marks



Q3

Red-Black Tree

4 marks



Q4

Red-Black Tree

4 marks



Q5

B-Tree

4 marks



Q6

Tree Terminology

4 marks



Q20 B-Tree

4 marks

In a B-Tree of order 4, what is the maximum number of children a node can have?

A. 2

B. 3

C. 4

D. 5

 Save Answer

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QUESTIONS

Q1Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6****Q21 Trie**

4 marks

In Trie, each node represents:

A. Prefix matching**B.** Arithmetic operations**C.** Graph traversal**D.** Sorting arrays Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6



Q22 Splay Tree

4 marks

Splay Tree is a type of:

A. Complete Binary Tree

B. Self-adjusting Binary Search Tree

C. Heap Tree

D. Graph

Save Answer

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QUESTIONS

Q1

Tree Traversal

4 marks



Q2

Binary Search Tree

4 marks



Q3

Red-Black Tree

4 marks



Q4

Red-Black Tree

4 marks



Q5

B-Tree

4 marks



Q23 Tree

4 marks

The height of a single-node tree is:

B. 1

C. -1

D. 2



Save Answer

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QUESTIONS

Q1

Tree Traversal
4 marks



Q2

Binary Search Tree
4 marks



Q3

Red-Black Tree
4 marks



Q4

Red-Black Tree
4 marks



Q5

B-Tree
4 marks



Q6

Tree Terminology
4 marks



Q24 Splay Tree

4 marks

Which rotation is used when the node is the direct child of the root?

20

/

10

A. Zig

B. Zig-Zig

C. Zag-Zig

D. Split

 Save Answer

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CO3_AT3_MCQ[← Back](#)**QUESTIONS****Q1**Tree Traversal
4 marks**Q2**Binary Search Tree
4 marks**Q3**Red-Black Tree
4 marks**Q4**Red-Black Tree
4 marks**Q5**B-Tree
4 marks**Q6****Q25 Splay Tree**

4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf**B.** Middle**C.** Root**D.** NULL Save Answer